SIAGA: Technical Validation Report

10Alytics Global Hackathon 2026, Track B (Data Science): Public Health AI Engineer ASEAN Digital Health and Climate Resilience Initiative (ADHCRI)

Executive summary

SIAGA (Indonesian and Malay for "alert, ready, on standby") is the predictive intelligence layer of the ADHCRI platform. It converts a fragmented set of ASEAN health indicators into three decision tools: a validated life-expectancy model, a ranked list of the drivers a health ministry can act on, and five-year forecasts of the region's communicable disease burden.

Headline results:

Question the model answers	Metric	Result
Predict the near future of a known country	MAE (2012 to 2014 held out)	**1.13 years**, $R2 = 0.92$
Generalize to a completely unseen country	MAE (leave-one-country-out)	**2.45 years**, $R2 = 0.71$
Hold up on independent external data	MAE vs World Bank 2015 to 2019	**2.08 years**, $R2 = 0.75$

The out-of-source test is the credibility claim: the model was trained only on the supplied 2004 to 2014 data and then scored against World Bank records for 2015 to 2019, a different source and a future time window it never saw.

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1. Problem and SDG alignment

ASEAN's ten member states span a life-expectancy range of roughly 16 years and a maternal-mortality range of more than 50 to 1. Reacting to health crises after they develop increases both mortality and cost. SIAGA supports a preventative posture by quantifying where the next health investment produces the most benefit.

- **SDG 3 (Good Health and Well-being):** the target variable and the disease forecasts address life expectancy and communicable disease burden directly.
- **SDG 10 (Reduced Inequalities):** Section 6 quantifies cross-country health gaps.
- **SDG 17 (Partnerships for the Goals):** the harmonized panel is published through an interoperability API so member states share one comparable schema.

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2. Data

Source: the ASEAN Public Health Task Force supplied 20 indicator files covering the ten member states, mostly 2004 to 2014 (references 1 and 2). One file was a verified duplicate and one was source metadata, leaving 18 usable indicators.

The panel is small and honest about it: ten countries by eleven years is at most 110 country-year observations. Every methodological choice below is made with that constraint in mind, and the validation strategy is designed to detect overfitting rather than hide it.

2.1 Data quality issues and resolution

All issues were found during exploration and resolved explicitly. No file was discarded without a stated reason, and no value was silently overwritten.

Issue	Example	Resolution
Country names differ across files	"Lao's PDR", "Lao PDR", "Lao People's Democratic Republic"; "Brunnei" typo	Mapped every variant to one canonical name and ISO3 code
Broken header in the target file	Two "2004M" columns in life expectancy	Second column identified as female and renamed
Numbers stored as text	HIV deaths as " 1 800" (space thousands separators)	Parsed with separator stripping
Year columns carry suffixes	"2010TBC", "2014Measles", "2004DPT"	Suffixes stripped during reshaping
Isolated data-entry errors	Cambodia TB 2010 = 125 amid values near 700	Detected as local spikes, adjudicated (Section 3), repaired by interpolation

2.2 Coverage-driven roles

Each indicator was assigned a role by its coverage in the modeling window:

- **Model features (well covered):** immunization (measles 99 percent, DPT 83 percent), government health spend (91 percent), undernourishment (90 percent), crude birth rate (88 percent), TB prevalence (68 percent), staffing densities (41 to 55 percent).
- **Model target:** life expectancy (89 percent), averaged across sexes.
- **Forecast targets:** TB and malaria prevalence.
- **Equity layer (context, not features):** maternal, infant, and under-five mortality.

3. Cleaning validated against an independent source

Suspected data-entry errors were not deleted on suspicion alone. A value was flagged only when it formed a local spike (more than three times away from both temporal neighbours), then adjudicated against the World Bank series for the same country and disease (reference 1). For example, the raw file records a collapse in Cambodian TB in 2010 (125 against neighbours near 700). The World Bank shows a smooth decline across those years with no collapse, confirming the raw value as an entry error. It was therefore repaired by within-country interpolation rather than left as a hole in a genuine trend.

Cross-source agreement confirms the harmonization more broadly:

Indicator	Correlation with World Bank	Median absolute difference
Life expectancy	0.93	1.18 years
Physicians per 1,000	0.95	0.02

Nurses and midwives per 1,000	0.98	0.03
Measles immunization	0.88	0.50 points
Undernourishment	0.90	1.90 points

4. Feature engineering

Three engineered decisions carry most of the value:

1. **Per-capita log health spend.** Absolute capital expenditure in million USD is not comparable across countries of vastly different size. We divide by World Bank population and take the natural log, reflecting the diminishing returns of spend.
2. **Contextual determinants: GDP per capita and sanitation access.** These are two of the strongest established determinants of life expectancy (the Preston curve for income; clean water and sanitation for disease transmission). Adding them from the World Bank lifted the out-of-source MAE from 2.65 to 1.96 years and the R2 from 0.54 to 0.76. Sanitation is also an actionable infrastructure lever, and SHAP ranks it as the single largest driver, consistent with the historical public-health record.
3. **HIV snapshot interpolation.** HIV prevalence is reported only in 2000, 2005, 2010, and 2016. Because population prevalence moves slowly, linear interpolation between snapshots yields a defensible annual feature.

Interior gaps are linearly interpolated within each country (never at the series edges, never on the target). Singapore's undernourishment, reported by the FAO as below the 2.5 percent floor rather than as a number, is set to that floor.

4.1 Leakage control

Mortality indicators (infant, under-five, maternal, crude death) were deliberately excluded from the feature set. Life expectancy is computed from mortality tables, so these variables are near-duplicates of the target (pooled correlations of -0.75 to -0.87). Including them would inflate the score while making the model useless for policy, since a ministry cannot budget "lower infant mortality" as an input. The retained features are all actionable or exogenous drivers.

5. Modeling

Two complementary models were fit.

Interpretable model. A linear mixed-effects panel model with country random intercepts produces standardized coefficients a policymaker can read directly. It is fit on complete cases only (30 of 98 rows), so its coefficients are read directionally and cross-checked against the machine-learning model rather than taken as precise effect sizes.

Predictive model. A histogram-based gradient-boosting regressor with two properties suited to this problem:

1. **Native handling of missing values,** so no fabricated imputation is forced onto the sparse staffing indicators.

2. **Monotonicity constraints** encoding medical priors: more physicians, nurses, pharmacists, immunization, or spend can never lower predicted life expectancy, and more TB, HIV, or undernourishment can never raise it. This guarantees the policy simulator cannot produce a perverse recommendation.

6. Validation

Three tests, each answering a different question. Reporting all three is the honesty argument, since any single number can flatter a small-sample model.

Test	Question	R2	RMSE	MAE
Leave-one-country-out	Predict a country never seen in training	0.70	3.03	2.68
Temporal split (train 2004 to 2011, test 2012 to 2014)	Predict the near future of known countries	0.93	1.43	1.08
Out-of-source, out-of-time (predict 2015 to 2019, score vs World Bank)	Hold up on external data	0.76	2.41	1.96

The temporal split matches the product's real use case (advising countries for which history exists) and is where accuracy is highest. The out-of-source test is the strongest external evidence: features for 2015 to 2019 were drawn from the World Bank, mapped into the model's schema, and scored against World Bank life-expectancy actuals the model never touched.

7. Driver isolation (explainability)

SHAP decomposes each prediction into per-feature contributions in years of life expectancy (reference 3). Across ASEAN, the largest lever is basic sanitation access (about 1.9 years of mean absolute impact), followed by communicable disease burden (TB about 1.0, HIV about 0.6), GDP per capita (about 0.9), physicians (about 0.6), crude birth rate (about 0.5), measles immunization (about 0.5), and health staffing. Sanitation leading is consistent with the historical public-health record, in which clean water and sanitation are among the strongest determinants of population life expectancy. This ranking is both the feature-importance deliverable and the engine behind the interactive policy simulator.

8. Epidemiological forecasting

The brief requires a five-year forward forecast of TB or malaria. We forecast TB per country (ASEAN carries roughly 45 percent of the global TB burden) and add malaria as a bonus where coverage allows.

Method: damped-trend exponential smoothing fit in log space, so forecasts stay positive, with 80 percent prediction intervals (reference 4). Each country is backtested by holding out its last three observed years and comparing the model against a naive last-value baseline. Where the naive baseline wins (4 of 10 countries for TB), the naive forecast is used for that country and labelled as such, rather than presenting a more sophisticated model that does not actually forecast better.

9. Limitations

1. **Small panel.** Ten countries and eleven years is a small sample. Mitigated by leave-one-country-out and out-of-source validation, and by reporting ranges rather than false precision.
2. **Mixed-effects coefficients** rest on 30 complete rows and are read directionally.
3. **Lao PDR's TB record is about 73 percent incomplete,** so its forecast carries wide intervals and is labelled low-confidence in the dashboard.
4. **The API's linear surrogate** is a sign-constrained approximation of the reference gradient-boosting model for lightweight edge inference (in-sample R2 = 0.94), not the primary model.

10. System architecture

- **Training pipeline:** a reproducible, versioned pipeline (`pipeline/`) run with `python -m pipeline.train`. It ingests raw files and World Bank data, cleans, engineers features, trains, validates, explains, forecasts, versions the model to `models/`, and refreshes the serving files. Retraining on new data is one command.
- **Model registry:** `models/` holds the deployed model, a model card, the feature spec, metrics, and `registry.json` (a metric history across versions), so model lineage is tracked and a regression can be caught or rolled back.
- **Analytics layer:** a reproducible Jupyter notebook (`notebooks/siaga_analysis.ipynb`) presenting the same analysis as a narrative from raw files to validated models.
- **Interoperability layer (SDG 17):** a dependency-free Go API (`api/`) serving the harmonized panel, forecasts, driver ranking, and edge predictions as a small static binary. It enforces API-key authentication and per-client token-bucket rate limiting, and ships as a distroless Docker image suitable for low-resource deployment.
- **Decision layer:** a Streamlit dashboard (`dashboard/`) with a regional map, an inequality lens, the SHAP driver view, an interactive policy simulator, and the disease forecasts.
- **Deployment:** Docker images and a compose file (`deploy/`) run the API and dashboard together. See `real_world_readiness.md` for an honest assessment of what is production-grade today and what a pilot would harden next.

Phase 2 roadmap

- Blockchain consent and audit layer for tamper-evident cross-border data sharing.
- Flutter offline-first field application for rural health workers, syncing to the API when connectivity returns.
- Integration of climate and mobility data for outbreak nowcasting.

References

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